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Facial Expression Recognition

Image Processing

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■ PROBLEM DEFINITION AND OBJECTIVES

Problem Statement

Automatic facial expression recognition is the task of mapping a facial image to one of a discrete set of emotion categories. It is a fundamentally difficult problem due to high intra-class variation (the same emotion expressed differently across individuals), low inter-class variation (visually similar expressions for different emotions), and dataset-level challenges including label noise, pose variation, uncontrolled lighting, and heavy class imbalance.

FER2013 encapsulates these challenges: images are 48×48 pixel grayscale photographs collected in the wild via automated web search, and the dataset is known for noisy labels and a significant imbalance between the Disgust class (436 training samples) and the Happy class (7,215 training samples).

Objectives

- Build a complete end-to-end FER pipeline from raw images to real-time inference.
- Design a preprocessing module that improves image quality without overfitting to the train set.
- Extract discriminative hand-crafted features that generalise well to unseen expressions.
- Train a regularised classifier that maximises validation accuracy and avoids overfitting.
- Integrate all modules into a single orchestrator notebook with one-click execution.
- Demonstrate real-time inference on static images and live webcam input.
- Document methodology, findings, and debugging decisions for reproducibility.

Scope and Constraints

- Dataset: FER2013 only. No external data or pre-trained deep features used.
- Features: hand-crafted descriptors (HOG). No convolutional feature learning.
- Classifier: fully-connected MLP. CNNs are excluded from the primary pipeline.
- Input modality: grayscale images. No depth, thermal, or multi-modal data.

■ METHODOLOGY

3.1 Dataset

FER2013 contains 35,887 labeled grayscale images across seven emotion classes. The official split provides 28,709 training samples and 7,178 test samples. Class distribution is highly imbalanced:

Class	Train	Test	Share
Angry	3,995	958	14.0%
Disgust	436	111	1.5%
Fear	4,097	1,024	14.3%

Class	Train	Test	Share
Happy	7,215	1,774	25.2%
Neutral	4,965	1,233	17.3%
Sad	4,830	1,247	16.9%
Surprise	3,171	831	11.1%

3.2 Preprocessing

Raw images are processed through a two-stage pipeline before feature extraction. All operations are applied to grayscale images only.

- CLAHE (Contrast Limited Adaptive Histogram Equalization): locally boosts contrast in a tile-grid (8×8) without amplifying sensor noise globally. Clip limit = 2.0.
- Unsharp masking: subtracts a Gaussian-blurred copy of the image to enhance high-frequency edge detail, making facial landmarks more salient for HOG.

Processed images are written to disk and used as the sole input to all downstream modules, ensuring that preprocessing decisions are decoupled from feature extraction.

3.3 Face Region Detection

Each preprocessed image is passed through MediaPipe BlazeFace (short-range model) to localise and crop the face region before feature extraction. Restricting HOG computation to the face eliminates background gradients that carry no emotion-relevant information.

- Model: blaze_face_short_range.tflite
- The grayscale image is temporarily converted to RGB for MediaPipe compatibility, then the detected bounding box is applied to the original grayscale.
- Fallback: if no detection is returned, the full image is used rather than discarding the sample.
- Output: face crop resized to 64×64 pixels.

3.4 Feature Extraction — HOG

Histogram of Oriented Gradients (HOG) encodes the local distribution of gradient magnitudes and orientations in a densely-sampled grid of cells. This representation captures the shape-based cues that distinguish facial expressions the curvature of a smile, the depression of brows in anger, the widening of eyes in surprise without requiring learned parameters.

Parameter	Value	Rationale
Image size	64×64	Richer gradient field vs native 48×48; no dimensionality explosion
Orientations	9	Covers 0–180° in 20° bins; standard for face tasks
Pixels per cell	(8, 8)	Balances spatial resolution with shift robustness
Cells per block	(2, 2)	Local contrast normalisation window
Block norm	L2-Hys	Clips dominant gradients; stable under illumination change
Transform sqrt	True	Gamma compression; reduces large-gradient dominance
Output dim	324	Flat float32 vector per image

An earlier design combined HOG with Local Binary Pattern (LBP, 10-bin) and Canny edge orientation histograms (18-bin), yielding a 928-dimensional vector. Empirical evaluation showed that the combined descriptor produced lower validation accuracy than HOG alone: the additional dimensions introduced noise that caused the MLP to overfit training data rather than learning generalisable structure. The pipeline was simplified to HOG only.

3.5 Feature Scaling

HOG feature vectors are standardised using StandardScaler (zero mean, unit variance) before being passed to the MLP. Neural networks are sensitive to input scale — unscaled features produce unstable gradients and slow convergence. The scaler is fitted exclusively on the training split and applied identically to the test split, preventing any leakage of test statistics into the training process.

- Train post-scaling: mean ≈ 0.000, std ≈ 1.000
- Test post-scaling: mean ≈ 0.0006, std ≈ 0.999
- Scaler saved to disk (scaler.pkl) and reloaded during inference to guarantee consistency.

3.6 Classifier — MLP Architecture

A fully-connected Multi-Layer Perceptron with three hidden layers is trained on the scaled HOG features. Batch Normalisation after each of the first two dense layers stabilises gradient flow and accelerates convergence. Dropout at decreasing rates (0.4 → 0.3 → 0.2) provides progressive regularisation, with heavier dropout at early layers where overfitting risk is highest.

Layer	Output	Regularisation
Dense (ReLU)	512	BatchNorm + Dropout 0.4
Dense (ReLU)	256	BatchNorm + Dropout 0.3
Dense (ReLU)	128	Dropout 0.2
Dense (Softmax)	7	—

- Optimizer: Adam, initial learning rate = 0.001
- Loss: sparse categorical cross-entropy
- Batch size: 64 | Max epochs: 100

3.7 Training Protocol

Before training, the dataset is split into a training portion (85%) and a validation portion (15%) using a stratified split (sklearn train_test_split, stratify=y_train, random_state=42). Stratification ensures all seven emotion classes are proportionally represented in the validation set — a critical requirement given FER2013's class imbalance.

Two callbacks govern the training loop:

- EarlyStopping: monitors val_loss with patience = 10 epochs. Training halts when no improvement is observed for 10 consecutive epochs, and the best weights are restored automatically.
- ReduceLROnPlateau: halves the learning rate when val_loss does not improve for 5 epochs, with a minimum floor of 1e-6. This allows fine-grained convergence without manual schedule tuning.

Note: Using Keras' built-in validation_split without prior shuffling was found to produce a near-single-class validation set (due to data being sorted by class folder), causing val_accuracy to collapse to ~8%. The stratified explicit split fully resolved this issue, restoring val_accuracy to the expected range from epoch 1.